

Biochemistry

1- The general formula of amines is

- ✓ R-NH₂
- R₂CH
- R-COOH
- R-CH₂

2- Thiamine deficiency leads to_____?

- Scurvy
- Rickets
- Pellagra
- ✓ Beri-beri

3- Regarding protein synthesis in eukaryotes, all proteins are initially synthesized with?

- ✓ Methionine at their N-terminus
- A multiple of 3 amino acids in their sequence
- Methionine at their C-terminus
- Tryptophan at their C-terminus

4- When double-stranded DNA is heated at neutral pH, which change does not occur?

- The helical structure unwinds
- The absorption of ultraviolet (260 nm) light increases
- ✓ The covalent N-glycosidic bond between the base and the pentose breaks
- The hydrogen bonds between A and T break

5- A non-competitive inhibitor of an enzyme ...

- Increases both K_m and V_{max}
- Increases K_m with no change in V_{max}
- Decreases both V_{max} and K_m
- ✓ Decreases V_{max} with no change in K_m

6- Which of the following vitamins provides the cofactor for transamination of amino acids?

Retinol

Niacin

✓ Pyridoxine

Phylloquinone

7- Tetracycline prevents synthesis of polypeptide by:

✓ Preventing binding of aminoacyl tRNA

Blocking mRNA formation from DNA

Releasing peptides from mRNA-tRNA complex

Competing with mRNA for ribosomal binding sites

8- The catalytic efficiency of two different enzymes can be compared by their

Molecular size of the enzymes

Product type

✓ Km values

Optimum pH value

9- In base-excision repair, the first enzyme to act is

✓ DNA glycosylase

DNA ligase

Dam methylase

DNA polymerase

10- A buffer solution contains propionic acid and its conjugate base; the pK_a of propionic acid is 4.88. At what pH would the solution buffer best?

✓ 5.0

4.0

6.0

3.0

11- Which vitamin requires cobalt for its activity?

Vitamin B2

Vitamin A

Vitamin D

✓ Vitamin B12

12- Which of the following can cancel the effect of an irreversible inhibitor?

Increasing the product concentration

Increasing the temperature

Increasing the substrate concentration

✓ Removing the inhibitor

13- Sphingosine is not a component of....

✓ Cardiolipin

Ceramide

Gangliosides

Sphingomyelin

14- If you replace the oxygen of the hydroxyl group with sulfur in R-OH, which functional group will you get?

hydroxyl

Sulfa

Sulfoxyl

✓ Thiol

15- Which of the following has more than one type of sugar in its structure?

Amylopectin

Inulin

✓ Hyluronic acid

Glycogen

16- In bacteria the elongation stage of protein synthesis does not involve

Peptidyl transferase

✓ IF-2

GTP

EF-Tu

17- Which vitamin is a major lipid-soluble antioxidant in cell membrane?

Vitamin K

Vitamin D

Vitamin C

✓ Vitamin E

18- Which of the following statements concerning fatty acids is correct?

They are strongly hydrophilic

✓ One of them is a precursor of prostaglandins

They all contain one or more double bond

They are a constituent of sterol

19- Non-steroidal anti-inflammatory drugs (NSAIDS) like aspirin and ibuprofen act by blocking the production of

Sphingolipids

Biological waxes

Vitamin D

✓ Prostaglandins

20- In sickle cell anemia, the basis of the malfunction of the hemoglobin molecule is ...

Reduced affinity for oxygen

✓ Substitution of a single amino acid

Incorrect secondary structure

Faulty binding of the heme groups

21- Polyprotic acids such as H_3PO_4 , can act as acid-base buffers....

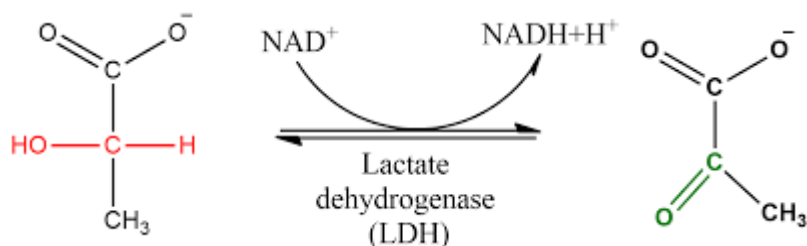
At pH values around neutrality

Only in combination with polyprotic bases

✓ At pH values around any of their pKa's

If their concentration is kept low

22- Which class of enzymes may catalyze the following reaction?



✓ Oxidoreductases

Ligases

Hydrolases

Isomerases

23- In polysaccharides, monosaccharides are joined by:

Ester bonds

✓ Glycosidic bonds

Peptide bonds

Glucose bonds

24- Dipalmitoyl lecithin acts as

✓ Lung surfactant

Platelet activating factor

Anti-ketogenic compound

Second messenger for hormones

25- Which of the following contains an ether-linked alkyl group?

Phosphatidyl serine

Gangliosides

Sphingomyelin

✓ Platelet-activating factor

26- Ciprofloxacin inhibits.....

DNA polymerase III



DNA gyrase

DNA topoisomerase I

DNA polymerase I

27- As the pKa of an acid increase, the acid will be:

Stronger

Converted to basic

Converted to neutral



Weaker

28- The compound that consists of ribose linked by an N-glycosidic bond to N-9 of adenine is



Adenosine

A purine nucleotide

Adenosine monophosphate

A deoxyribonucleoside

29- The tryptophan operon of E. coli is repressed by tryptophan added to the growth medium. The tryptophan repressor probably



Binds to the trp operator in the presence of tryptophan

Binds to RNA polymerase when tryptophan is present

Is an attenuator

Binds To The Trp Operator In The Absence Of Tryptophan

30- The lipoprotein richest in cholesterol is ...

VLDL



LDL

HDL

Chylomicron

31- The operator region normally can be bound by

- ✓ Repressor
- mRNA
- Inducer
- Suppressor tRNA

32- Which one of the following results in Ehlers-Danlos syndrome?

- Deficiency of vitamin C
- Deficiency of elastase
- ✓ Deficiency of lysyl hydroxylase
- Deficiency of antitrypsin

33- Histones are _____ that are usually associated with _____.

- Basic proteins -RNA
- ✓ Basic proteins - DNA
- Acidic proteins -RNA
- Acidic proteins -DNA

34- The secondary structure of protein is held together by which type of bonding?

- Covalent bonding
- Spatial arrangement
- 3-D stereochemical folding
- ✓ Hydrogen bonding

35- When a DNA molecule is described as replicating bidirectionally, it means that it has two ...

- ✓ Replication forks
- Origins
- Termination points
- Independently replicating segment

36- An amino acid not found in proteins is ...

Proline

✓ β -Alanine

Lysine

Histidine

37- Which one of the following antibiotics does not function by interfering with the translational process?

✓ Penicillin

Puromycin

Streptomycin

Cycloheximide

38- Which is incorrect statement about vitamin C?

It is involved in hydroxylation of prolyl- and lysyl- residues of collagen

It shows antioxidant activity

✓ It can be synthesized in the body

It is a strong reducing agent

39- The gene of lac operon which has constitutive expression is.....

✓ i

p

z

c

40- The proofreading function of DNA polymerase involves all of the following except

A 3' → 5' exonuclease

Detection of mismatched base pairs

Base pairing

✓ Reversal of the polymerization reaction

41- Deficiency of which one of these vitamins is the major cause of blindness?

Vitamin B12

Vitamin B6

✓ Vitamin A

Vitamin D

42- Which of the followings inhibit the DNA-dependent RNA polymerase in prokaryotes?

✓ Rifampicin

cycloheximide

Chloramphenicol

Ciprofloxacin

43- When $K_m = S$, reaction velocity (v) would be equal to.....

✓ $1/2 V_{max}$

$2 V_{max}$

$1 V_{max}$

$3/4 V_{max}$

44- Glycine and proline are the most abundant amino acids in the structure of

Myoglobin

Hemoglobin

Insulin

✓ Collagen

45- All of the following vitamins provide the cofactor for pyruvate dehydrogenase, except...

Thiamine

✓ Folate

Riboflavin

Niacin

46- Which of the following is a pair of non-reducing carbohydrate?

Cellobiose and sucrose

Lactose and maltose

Trehalose and maltose



Sucrose and trehalose

47- Which of the following would produce Methemoglobinemia?

Deletion of one β -globin locus

Deletion of one α -globin locus

Transition of Glutamic acid to Valine



Converting Ferrous iron (Fe^{2+}) to Ferric (Fe^{3+})

48- In scurvy, which amino acid that is normally a part of collagen is not synthesized?

Hydroxytyrosine



Hydroxyproline

Hydroxytryptophan

Hydroxyalanine

49- Which of the following disaccharides contains a beta-glycosidic bond?

Maltose and amylose



Lactose and cellobiose

Lactose and sucrose

Maltose and sucrose

50- RNA polymerase from E. coli (core enzyme alone) has all of the following properties except that it



Recognizes specific start signals in DNA

Is required for the synthesis of mrna, rrna, and trna in E. Coli

Can extend an RNA chain and initiate a new chain

Produces an RNA polymer that begins with a 5'-triphosphate

51- All of the following statements correctly describes transport of O₂ by hemoglobin except

The plot of percentage of O₂ bound versus O₂ pressure is sigmoidal in shape

- ✓ Increased CO₂ concentrations increase O₂ affinity
- All of the four heme moieties binds O₂ cooperatively
- CO binds to hemoglobin more avidly than does O₂

52- Oligosaccharides linked to proteins are called:

- Glycolipids
- Glycosaminoglycans
- Glucosides
- ✓ Glycoproteins

53- Which one of the following is a reversible inhibitor of the COX-1 enzyme that is involved in prostaglandin synthesis?

- Penicillin
- Asprin
- ✓ Ibuprofen
- Sarin

54- Which of the following is not an example of a weak acid?

- ✓ Sulfuric acid
- Carbonic acid
- Pyruvic acid
- Lactic acid

55- Which one of the following is assembled and secreted by the liver?

- High Density Lipoprotein (HDL)
- ✓ Very Low-Density Lipoprotein (VLDL)
- Low Density Lipoprotein (LDL)
- Chylomicrons

56- The enzyme which hydrolyses Glucose-Fructose linkage is....

Amylase

Lactase

Maltase

✓ Sucrase

57- Deficiency of which one of these vitamins may lead to megaloblastic anemia?

Vitamin D

Vitamin E

Vitamin B6

✓ Vitamin B12

58- Which of the following fatty acids has the lowest melting point?

Palmitic acid

Stearic acid

✓ Eicosapentaenoic acid

Arachidonic acid

59- The strength of an organic acid depends on:

Number of double bonds

Number of neutrons gained

Number of hydroxyl released

✓ Electronegativity

60- Formic acid is also called

✓ Methanoic acid

Formaldehyde

Ethanoic acid

Acetone

61- Amylopectin has:

$\beta(1 \rightarrow 4)$ and $\alpha(1 \rightarrow 6)$ linkage

$\beta(1 \rightarrow 2)$ linkage

✓ $\alpha(1 \rightarrow 4)$ and $\alpha(1 \rightarrow 6)$ linkage

$\alpha(1 \rightarrow 2)$ linkage

62- Which of the following produces N-acetyl glucosamine upon hydrolysis?

✓ Chitin

Inulin

Dextran

Cellulose

63- Structural polysaccharides include:

Cellulose and starch

Glycogen and starch

Chitin and glycogen

✓ Cellulose and chitin

64- Choose the incorrect statement out of the following.

Glycine is optically inactive

Selenocysteine is 21st amino acid

✓ Tyrosine is a modified amino acid

Only L-amino acids are found in the biological system

Cell Physiology

65- Regarding the electrical potential, which of the following is correct?

It is the same as action potential.

It generates rapidly changing electrochemical impulses at all cell's membrane.

✓ It exists across the membranes of virtually all cells of the body

Amount of this potential is the same in all cells.

66- The main factor that makes plateau phase of an action potential in cardiomyocytes is

- opening of voltage-activated potassium channels
- closure of voltage-activated potassium channels
- opening of voltage-activated sodium channels
- ✓ opening of voltage-activated calcium-sodium channels

67- Which of the following is not of importance in osmosis process?

- Impermeability of the membrane to solutes
- Number of osmotic particles
- Concentration difference for water between 2 sides of membrane
- ✓ Mass of the particles

68- How does the resting membrane potential change in a nerve cell if extracellular K concentration begin to increase over than normal?

- Nothing will happen.
- It makes the membrane potential more negative.
- K leak channels begin to be inactivated.
- ✓ It shifts towards less negative potential.

69- In this type of communication, message transmission is performed directly from cell to cell.

- Endocrine
- Autocrine
- Synaptic
- ✓ Gap junctions

70- Which of the following is not of great importance in Nernst equation?

- ✓ Existence of uncharged molecules
- Existence of electrical gradient between 2 sides of the membrane
- Chemical gradient between 2 sides of the membrane
- Existence of membrane permeability

71- What does "internal environment" mean?

It includes just the fluid inside the blood vessels.

It refers to the fluid inside the cells.



It is the same as extracellular fluid.

It means the whole fluid in human body.

72- In this type of communication, the products of cells diffuse in the extracellular fluid to affect neighboring cells.

Neural



Paracrine

Autocrine

Endocrine

73- Which of the following can not take into account as a second messenger?



Guanylyl cyclase

Ca²⁺

Cyclic adenosine monophosphate

Inositol triphosphate

74- If extracellular concentration of Na ions falls below physiologic range



amplitude of depolarization phase decreases.

resting membrane potential significantly becomes unstable.

no change occurs.

repolarization phase decreases.

75- Carbohydrates taking part in the cell membrane structure

have equal distribution in both sides of the cells.



are found just in extracellular side of the membrane.

don't play a major role in homeostasis.

attach just to phospholipid molecules.

76- acts as a glucose sensor in β -cells of pancreas.

GLUT 6

GLUT 4

GLUT 1

✓ GLUT 2

77- What is the name of the process in which in the presence of a deficiency of the chemical messenger, there is an increase in the number of active receptors?

Receptor-mediated endocytosis

Sensitization

Down-regulation

✓ Up-regulation

78- Which of the following transport pathways dose not need a carrier protein?

✓ Simple diffusion

Facilitated diffusion

Primary active transport

Secondary active transport

Histology

79- Lipid Raft structure located in which part of the cell membrane?

Extracellular side

Peripheral proteins sides

✓ Integral proteins sides

Cytoplasmic side

80- Genetic disorders of intermediate filaments is most observed in which of the following structures?

✓ Nucleous

Glycogen

Cell Membrane

Proteasome

81- Synthesis of which organelle is occurred in the nucleolus?

RER



Ribosome

Peroxisome

Proteasom

82- In which of the following cases the protein not related to the relevant organelle?

Mitochondria-Purine

Peroxisome-catalase

Proteasome - Ubiquitin



RER - Clatrin

83- Pale, large and centrally located nucleus in the cell, shows:

This cell will differentiate

This cell is dead



This cell is active in protein synthesis

This cell starts apoptosis

84- All of the following are true about apoptosis except:

In apoptosis, intracellular proteases are activated

Apoptosis occurs in fetal development

In apoptosis, apoptotic bodies are formed



Apoptotic cells resulted in inflammation

85- Oligosaccharide chains that extend outward from the cell surface and contribute to a delicate cell surface coating called the:

multiadhesive glycoproteins

basal lamina



glycocalyx

glycosaminoglycans

86- Which of the following terms is applied to the smallest chromosome subunit in eukaryotic cells?

Chromosome

Euchromatin



Nucleosome

heterochromatin